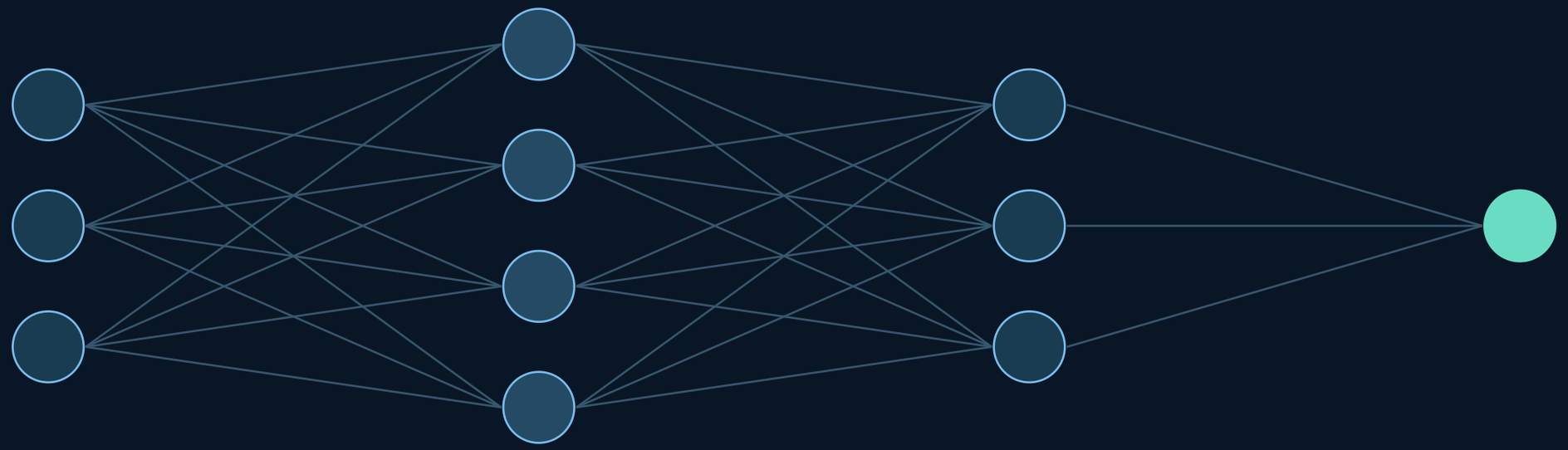
Inside a neural network

A visual guide to the six components that make learning possible

FORWARD PASS



FEATURES -> WEIGHTED SUMS -> NONLINEAR ACTIVATIONS -> PREDICTION

The connections represent learned weights; each circle represents a neuron.

The building blocks

Each component has a distinct role in the forward pass

STRUCTURE

01 Layers

Organize neurons into an input layer, one or more hidden layers, and an output layer. Depth allows progressively transformed representations.

COMPUTATION

02 Neurons

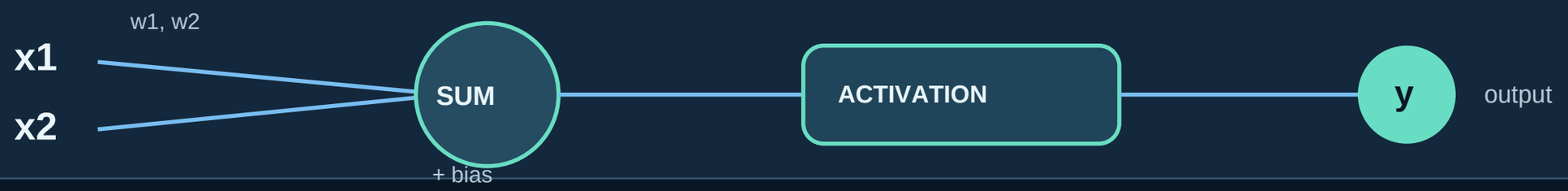
Combine incoming values with weights and a bias, then apply an activation function to create an output.

PARAMETERS

03 Weights

Trainable connection strengths that determine how strongly each input influences a neuron. Learning changes these values.

INSIDE ONE NEURON



Why activation matters

Without nonlinear activations, stacked linear layers remain a linear transformation

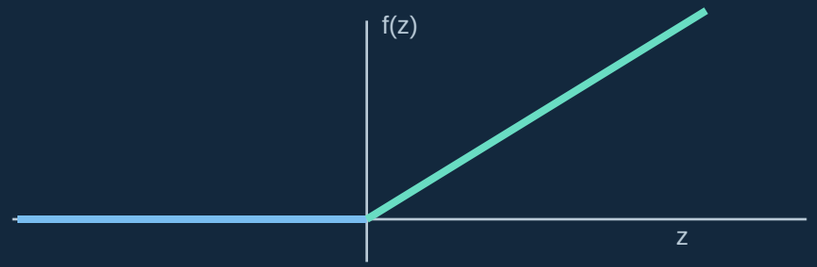
TRANSFORMATION

04 Activation functions

Turn the weighted sum into a neuron output. Common choices include ReLU, sigmoid, and tanh; their nonlinear shapes enable complex decision boundaries.

EXAMPLE / ReLU

f(z) = max(0, z)



WORKED CONCEPT

Inputs x = [2, 1] Weights w = [0.5, -1] Bias b = 0.5

Weighted sum z = (2 x 0.5) + (1 x -1) + 0.5 = 0.5 -> ReLU(z) = 0.5

Illustrative arithmetic only; not a trained-model result.

How a network learns

Predictions are evaluated, then parameters are adjusted to reduce error

ERROR SIGNAL

05 Loss functions

Quantify the mismatch between a prediction and its target. Examples: mean squared error for regression and cross-entropy for classification.

PARAMETER UPDATES

06 Optimization algorithms

Use gradients of the loss to update weights and biases. Gradient descent and Adam are common; learning rate controls update size.

1 **Predict** > 2 **Measure loss** > 3 **Backpropagate** > 4 **Update weights**

Repeat over training batches; check validation performance to detect overfitting.

From diagram to experiment

Reproducible Playground activities and why visualization matters

TENSORFLOW PLAYGROUND / TEST PLAN

- **A** Hold dataset fixed; compare 1 vs. 2 hidden layers.
- **B** Increase noise; observe boundary stability and train/test loss.
- **C** Toggle input features or activation; compare convergence and complexity.

KEY INSIGHT

Visualizing networks connects architecture, feature flow, and learning behavior. More neurons or layers can increase expressiveness, but do not guarantee better generalization; evaluate both training and test performance.

Exploration prompts are proposed, not measured findings. Record screenshots and values after running.

References: TensorFlow Playground (playground.tensorflow.org); Goodfellow et al., Deep Learning (2016).